

IoT-Based Real-Time Crowd Monitoring System

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Abstract—There is an increasing demand for intelligent infrastructure in the modern world. Also, there is a need for efficient, automated people-counting systems. Traditional systems face some limitations, such as high implementation costs and privacy concerns. To address these challenges, this study proposes an IoT-based people counting system that uses Wi-Fi signals to count active users inside a selected area. This specific system is built with ESP32 microcontrollers configured to capture Wi-Fi probe requests received from nearby devices (specifically laptops). Each captured frame will be filtered and make an exact count. This filtering phase will happen in the cloud platform. (A virtual machine works as a dedicated server for filtering purposes.) Then it will be displayed in a web dashboard.

The stored data is processed to remove duplicate MAC addresses, while vendor-based filtering is employed to distinguish between device types such as laptops and smartphones. This method makes a more accurate estimation of the actual number of people inside a selected zone. It can provide valuable insights about space management and occupancy analysis. However, it was observed that the MAC address randomization has been used in most modern devices these days. To mitigate this problem, there is a specific approach introduced in the research. Since this solution is limited to the library, the study assumes that every user uses a laptop inside the library. Since Laptops' MAC address randomization is turned off by default, a single laptop can be counted as a single user. Future work will focus on addressing limitations such as MAC address randomization and fluctuating signal strength by incorporating machine learning algorithms for pattern recognition and ground truth calibration to improve accuracy under different environmental conditions.

Overall, this study highlights the potential of Wi-Fi-based sensing is a cost-effective solution for indoor environments. The proposed model can be applied in a variety of contexts, such as libraries, classrooms, shopping malls, and public facilities, etc. Also, with future improvements, it can be used in open environments.

Keywords—component, formatting, style, styling, insert (key words)

I. INTRODUCTION (HEADING 1)

An IoT-based real-time crowd counting system is a smart counting system introduced as a solution for crowd control in public areas. Crowd management is an important aspect of

operational efficiency in many public areas, such as universities, hospitals, and shopping malls. With increasing population density, overcrowded environments lead to discomfort, reduced efficiency, and time wastage. Traditional methods like density estimation, regression-based counting, and manual methods are more complex, less effective, and labor-intensive. But the proposed solution is a simpler, more effective, and privacy-preserving method. The rapid increase in metropolitan expansion and growth in population has significantly increased congestion in public environments. Inefficient crowd management is a major concern for modern cities, airports, educational institutions, and event organizers. Traditional CCTV-based systems are often expensive, require high computational power. Such systems can transform how public spaces are managed by replacing manual monitoring with intelligent, data-driven approaches. This project remedies the shortcoming by considering a crowd estimation system for laptops since their transmitted MAC addresses are non-randomized. This project uses ESP32 microcontrollers for sniffing Wi-Fi probes and sending the obtained information to a cloud database. A virtual machine running on Microsoft Azure further processes this information to remove randomized MAC addresses and matches the actual MAC addresses with the IEEE OUI database to identify laptop users. The resulting filtered information is then presented to the students and staff using a real-time web interface.

This project started as a solution to control high crowd density in the university library. During exam periods, many students come to the library and are unable to find a place to study due to the high number of students inside the university. In most university environments, students have no reliable way to know how crowded a space is before entering it. Therefore, this IoT-based system is an affordable, scalable, and privacy-friendly solution that can provide real-time crowd data to both users and administrators. Users often lack access to real-time occupancy information, which leads to poor decision-making and frustration. While some crowd-monitoring systems exist, they typically depend on cameras or motion sensors that are limited by cost, power, and range. Furthermore, their accuracy can degrade in poor lighting or crowded environments where individuals overlap in view. By addressing this matter, we have come up with a solution that notifies students when a specific place is overcrowded and

provides a live feed of crowd levels inside those specific areas. As a result, students can avoid overcrowded areas in the university.

This solution mainly aims to reduce crowd congestion and improve user experience. Furthermore, this design includes two ESP32 boards, which can detect the number of people using Wi-Fi probe request sniffing. Afterward, data is transferred using Azure Functions and stored in the Azure SQL database. Eventually, we developed a web application dashboard to display live crowd levels or users. This system can be adapted to various environments.

This research is meaningful because overcrowding directly affects user comfort, safety, and efficiency in shared public places. The other implementations we described above are costly image processing or manual monitoring and are not very suitable for smaller entities. On the other hand, better crowd supervision can help guide safety regulation and emergency response planning. The program has a beneficial impact on sustainable space use by reducing unnecessary congestion and travel time. Real-time trend analysis can be carried out through continuous data capture for peak hours and pattern recognition. Predictive modeling can be used to predict future congestion using historical data. This information is beneficial for the administrators while planning capacity and allocating resources. The proposed solution offers low deployment cost, high flexibility, and privacy protection. This kind of system can help build a smart infrastructure and data-driven decision-making.

II. RELATED WORK

Crowd counting inside indoor environments has been a major topic area for significant research because of congestion avoidance, facility management, safety precautions, and resource optimization. Traditional methods for crowd monitoring include manual head counting, video surveillance, and IoT sensor-based monitoring. Out of those traditional methods, the manual head-counting method requires human labor and results in errors. Video and image-based crowd counting methods raise privacy concerns and require high-cost computational resources and complex data analytics. [1] [6] Increasing the crowd density makes those systems underperform. [7] [8]

To overcome these limitations, wireless signal-based methods have emerged as better alternatives. Those methods mainly focus on Wi-Fi signals and Bluetooth signals emitted by electronic mobile devices. Several studies have focused on detecting mobile devices by capturing Wi-Fi probe requests, which are known as Management Frames emitted during the device's scanning phase in search of an access point. [2] The main concept was to map the number of unique MAC addresses to count now. Those methods provide a low-cost, efficient, and privacy-conscious approach for occupancy estimation.

However, it is observed that modern smartphones randomize their MAC addresses (MAC address Randomization), by making it impossible to uniquely identify each mobile device through captured MAC addresses.[3] Several researchers have proposed approaches to address this problem, like doing ground truth calibration to optimize the counting algorithm and training machine learning models to correlate the obtained results to the actual crowd count.

Other solutional approaches, instead of traditional crowd counting methods, mainly include detecting and analyzing Bluetooth signals.[4] This method utilizes periodic advertisement packets emitted by mobile devices and other wearable devices in a fixed time interval. This method has offered a more accurate result in the detection of devices while consuming low power but has been limited to a very low number of Bluetooth-enabled devices in real-world scenarios.

Image and video surveillance-based crowd counting is the most explored method at the time of writing this research paper. Computer vision algorithms process the obtained live feeds to detect and count the crowd in real time. This approach gives highly accurate results, but under controlled conditions like constant lighting conditions and the presence of high computational power. This approach leverages ethical and privacy concerns. The complexity of the implementation and maintenance of these systems, and due to the number of computational resources it needs, limits its practical deployment in university libraries and indoor environments. Crowd detection using IoT sensors like heat sensors, motion detection, and pressure sensors is highly sensitive to environmental conditions like temperature, crowd behavior, noise levels, air quality, etc., which makes that method unreliable. [5]

In contrast to the above methods, the use of laptop detection for crowd estimation in indoor environments focuses on a specific aspect of device usage patterns. It allows for a more targeted and reliable crowd count analysis. Unlike other mobile devices and Apple laptops, all other laptops generally have the MAC address randomization feature turned off by default. That behavior makes them a reliable signal indicator of human presence in indoor environments where consistent laptop usage is present. It is found that none of the researchers focused on this characteristic to estimate the crowd size in an indoor environment where it consists of laptop users. Only we have implemented a complete end-to-end system that integrates real-time data capture, cloud-based processing, and a close estimation of crowd density.

The proposed system builds on the above insights by using ESP32-based Wi-Fi sniffers, IEEE-defined MAC randomization rules, and vendor-based OUI classification to filter laptop devices from other mobile devices. By continuously capturing probe request frames, this system displays a threshold-based alert on crowd count in an online dashboard. The integration of cloud services for storage, filtering, and real-time dashboard updates addresses some major issues of prior research conducted.

III. METHODOLOGY

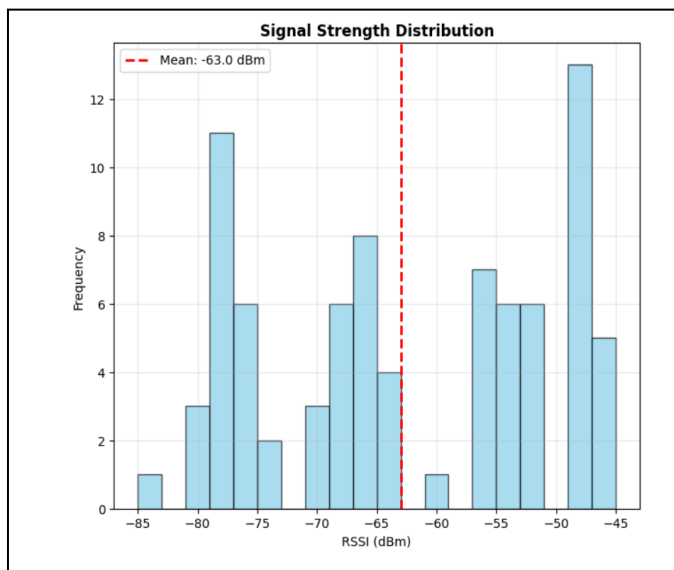
The methodology of this research focused on achieving a real-time crowd estimation within a university indoor environment using Wi-Fi probe request-based device detection. The overall process consisted of five main phases, starting from signal capturing, data transmission, cloud database storage, and processing the raw data to get a meaningful threshold-based alert. Each stage was systematically implemented and tested to ensure accuracy, reliability, scalability, and availability while preserving privacy and maintaining a low implementation cost.

A. System Overview

The implemented system integrates hardware, software, and cloud components to ensure smooth operation between the above-mentioned stages. At the hardware layer, first ESP32 microcontrollers were used as a Wi-Fi sniffer to capture Wi-Fi probe requests emitted by mobile devices in search of a Wi-Fi network within the library environment. These Wi-Fi probe requests mainly contain the MAC address and the RSSI value. The captured raw data is continuously transmitted to the second ESP32 board via UART connection in CSV format. Then the second ESP32 board converts the received data to a JSON format by adding a timestamp for each received data, and then accumulates all received data for 1 minute into a batch. Then the 1-minute batch is uploaded to the Azure Function endpoint over HTTPS, which eventually stores it in the cloud-hosted SQL database. The upper layer of the system architecture consists of an Azure Virtual Machine that fetches the raw data from the database and runs the filtering algorithm. The filtered data is uploaded to a new table in the same SQL database. Inside the VM, a web-based dashboard is hosted on a Nginx web server, which visualizes the crowd count as a threshold-based alert

B. Signal Capturing

The first ESP32 board is operating in promiscuous mode to capture Wi-Fi probe requests transmitted within range. The program running inside the board extracts only the relevant fields, which are the MAC address, RSSI value, and the timestamp. These captured raw data consist of the MAC addresses of all types of mobile devices, such as smartphones, laptops, smartwatches, etc. Most of these captured MAC addresses are randomized MAC addresses due to the MAC address randomization of new mobile devices. From further studies carried out, it was revealed that in laptops, MAC address randomization is turned off by default. This ESP32 device continuously captures Wi-Fi probe requests emitted by all mobile devices and temporarily stores them before transmitting to the second ESP32 board. [9]



C. Data Transmission

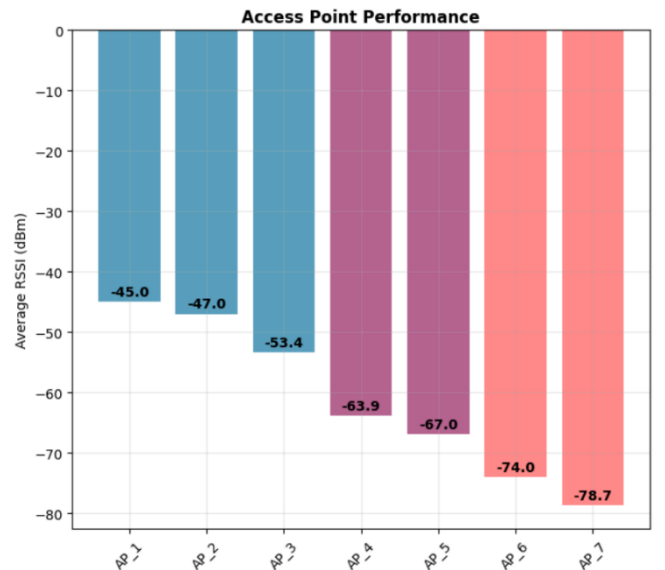
The captured raw data was sent to the second ESP32 board through the UART serial communication channel. This separation of workload among the ESP32 boards ensures the minimization of performance bottlenecks. The first board concentrates on packet sniffing, while the second board manages the sending of data to the cloud. The second board establishes a connection with a locally available Wi-Fi network and forwards the JSON payload to the Azure function app endpoint using the HTTP POST method. This process enables reliable, real-time data updates in the database. [11] [12]

D. Cloud Storage and Processing

Data is stored in the Azure SQL database, which serves as the data repository for both the filtering program and the dashboard backend. The Azure cloud environment was selected for its scalability, high availability, and other application support. The Python-based filtering program running inside the Azure virtual machine continuously processed the stored data to filter out the randomized MAC addresses and separate laptop MAC addresses. This filtering algorithm uses the IEEE-defined rules for MAC address randomization, which involve the locally administered bit of the MAC address to differentiate between real and randomized addresses. Local administered bit is set to 1 in Randomized MAC addresses. [14] To separate the laptops from other mobile devices, the filtered MAC addresses are checked against the laptop manufacturers defined in the Organizationally Unique Identifier (OUI) database. [13] Thereby, the final filtered output represents only the laptop MAC addresses. [15]

E. Environmental Calibration

One of the main problems this research faced was identifying the exact distance the Wi-Fi probe requests are coming from to filter out laptops from outside the library environment. Our approach was to consider the RSSI value to get an idea about the distance. But several tests conducted confirmed that there is no correlation between the RSSI value and the distance to the laptop from the ESP32 antenna.



F. Data Validation

Several manual headcounts were carried out at various times of the day, which represented both rush hours and the less crowded hours. Each recorded dataset was then compared with the final crowd count obtained through the website. Statistical analysis demonstrated an acceptable correlation between the detected laptop count and the manual headcount. It concluded that the project was successful, and the outliers, such as laptop sharing, individuals using multiple laptops, and internet-disconnected laptops, did not have much effect on the accuracy of the final output.

G. Data Visualization

A web dashboard hosted on an Nginx server inside the Azure VM fetches the final filtered laptop count from the Azure SQL database. This dashboard was developed using modern web technologies to provide both real-time and historical crowd counts. This user-friendly interface allows both the students and the staff to make an informed decision based on the crowd count displayed on the dashboard. This dashboard is accessible via a web browser on any mobile device.

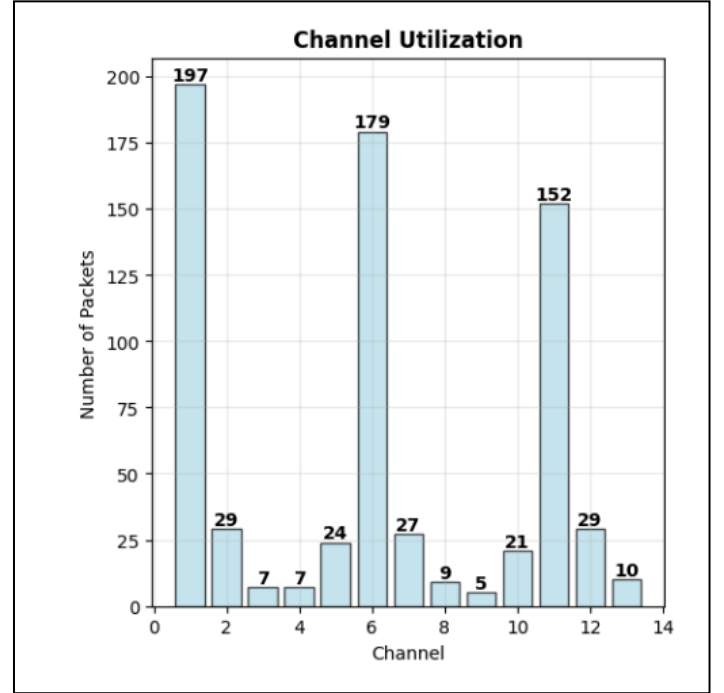
IV. RESULTS

The system was tested during several phases of the implementation. Those test scenarios included multiple real-world conditions inside the library. The target was to evaluate the working of each hardware component, the implementation of the whole system, and its accuracy, stability, and real-time responsiveness. Those results were compared with the manual headcounts to validate the assumption of correlation between the manual headcount and the number of internet-connected laptops or laptops that are actively searching for a network connection.

A. Data Collection

This evaluation was carried out over a period of one and a half months, which represented the critical university library's operating hours. One ESP32 board was employed to collect probe requests. In the initial phase ESP32 board was programmed specifically to focus on a few known MAC addresses. Then the laptops, which belonged to those known MAC addresses, were strategically placed in different locations inside the library. After several optimization changes to the program, it successfully detected all 4 MAC addresses with the details: MAC address, RSSI value, Frame

type, and the timestamp. Those frame types contained only the Wi-Fi probe requests. It was observed that the access points inside the library are operating on 5GHz. Laptops that were already connected to the network emitted probe requests at a very low rate. It was confirmed that the laptops have the feature “dual band operational capability” through the captured data, which contained 2GHz probe requests emitted by laptops that were already connected to a 5GHz access point. [10]



B. Filtering Performance

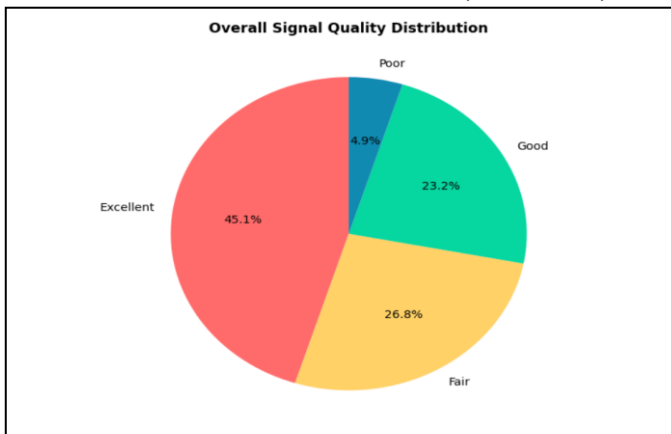
During the initial steps, the captured data contained randomized MAC addresses, which were mainly generated by mobile phones. It was observed that Samsung M13 emitted nearly 7 probe requests and Apple 16 emitted nearly 14 probe requests. All those probe requests contained randomized MAC addresses. The filtering algorithm, which used the IEEE-defined rules for MAC address randomization, successfully filtered out over 90% randomized MAC addresses and separated the real MAC addresses.

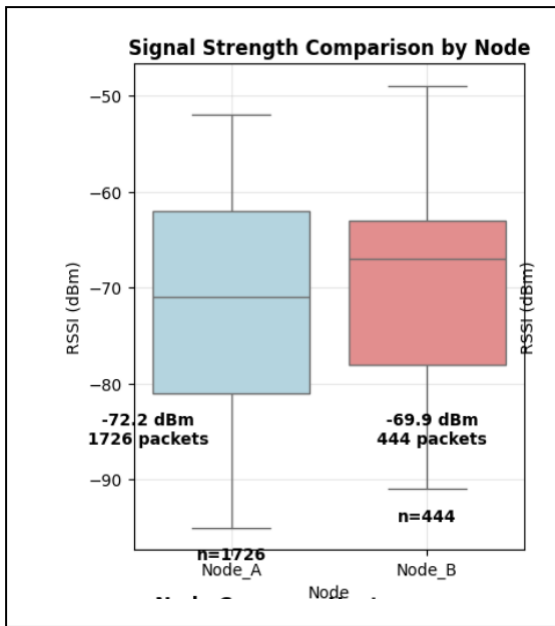
Out of those real MAC addresses, addresses that represented the laptops were filtered by integrating the OUI-based vendor details and doing a comparison.

These combined raw data filtering stages increased the accuracy of the information given to the students and staff via the online dashboard.

C. Accuracy Validation

The direct comparison between the filtered laptop counts and manual headcounts displayed an acceptable correlation.





preserving alternative for crowd count estimation in indoor environments where network-connected laptops dominate the

Total Packets	696
Unique Devices	114
RSSI Range	-92 to -36 dBm
Avg RSSI	-73.5 dBm
Channels Used	13
Most Active Channel	Ch 1
Best Signal	-36 dBm
Worst Signal	-92 dBm

mobile device presence.

H. Sample Result Set

D. Latency and Responsiveness

The end-to-end system latency was approximately 90 seconds between packet capturing to visualizing the crowd count in the online web dashboard. Azure cloud integration has positively affected this much performance, which ensured the near real-time performance for crowd monitoring. The dashboard automatically refreshed every 10 seconds to provide an updated view of the crowd level.

E. Reliability and Stability

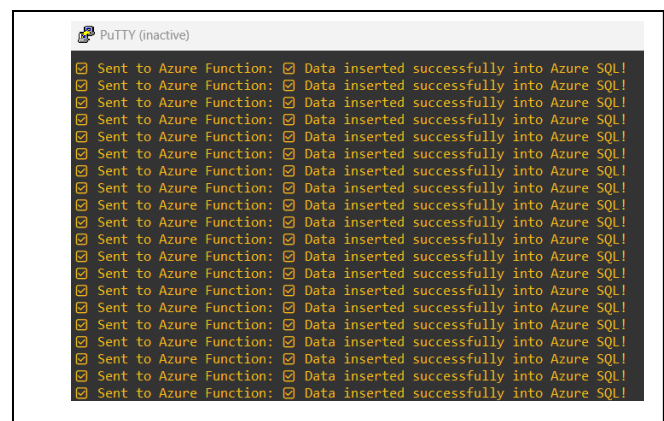
To maintain the stability of the hardware system, ventilation mechanisms and continuous, uninterrupted power supplies are provided. Throughout the testing period, both ESP boards operated successfully with no major disruptions. Occasional packet loss was observed during high network traffic periods, but since the sniffing operation was running continuously, it was able to detect those lost packets in the next sniffing rounds. The backup power supply, which was provided by a battery, was able to operate the system for 2 hours. That 2-hour period is more than enough since the university premises already have the backup generators running 24x7.

F. Real-Time Visualization of Results

An online web dashboard hosted through the Nginx web server inside the Azure VM is used to visualize the final processed data as a real-time crowd level. This informative visualization transformed the raw captured data into actionable information for both staff and students. That enabled the main goal of this initiative, which was the effective resource utilization, crowd management, facility management, and eventually increasing the efficiency of both university staff and the students.

G. Summary of Findings

The experimental results demonstrated the efficiency of this new approach, its specificity for university indoor environments. These results confirmed the feasibility of using Wi-Fi probe requests-based device detection specifically for laptops as a cost-effective and privacy-



id	mac	rssi
27723	70CF497118F3	-65
27724	48E7DA8B86DB	-74
27725	8C035858708E	-79
27726	2AEF40987C43	-68
27727	8E1DFA12CD88	-80
27728	1A5CDE781234	-77
27729	32DDAA4C5690	-72
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I. Pseudocode version of the two-stage MAC address filtering process

Algorithm TwoStageMACFilter

Input: raw_mac_list, oui_database

Output: filtered_mac_list

Initialize filtered_mac_list = []

For each mac in raw_mac_list:

If is_valid_format(mac):

first_byte = hex_to_int(mac[0:2])

Stage 1: Bit-level filtering

If (first_byte & 0b00000010) == 0:

prefix = mac[0:8].upper()

Stage 2: OUI verification

If prefix in oui_database:

filtered_mac_list.append(mac)

Return filtered_mac_list

Results Messages				
Search to filter items...				
id	mac	vendor	observed_time	rssi
106	70CF497118F3	Intel Corporate	2025-10-26T08:45:00.0000000	65
107	48E7DA8B86DB	AzureWave Technology Inc	2025-10-26T08:45:00.0000000	67
108	8C035858708E	Intel Corporate	2025-10-26T08:45:00.0000000	62
109	985F418D4203	Intel Corporate	2025-10-26T08:45:00.0000000	70
110	F44EE3A33266	Intel Corporate	2025-10-26T08:45:00.0000000	64

V. DISCUSSION

The implementation of the proposed IoT-based people counting system demonstrated the potential of wifi signal analysis as a real-time, practical crowd estimation. The use of ESP32 microcontrollers was cost-effective for the project implementation. The system captured probe requests without needing any additional hardware. The system collected a substantial amount of data, which was transmitted to the

cloud database and displayed on a real-time web dashboard. With this approach, students can see the crowd levels of selected zones in the university before they come to the university.

During the testing phase, there were some difficulties with RSSI values varying significantly with distance, device orientation, and environmental obstacles. Also, there was a big issue with MAC address randomization. To address these obstacles, this study has focused on vendor-based filtering, and as a future improvement, it can use ground truth calibration with machine learning.

From a network engineering perspective, the system integration with Azure cloud services ensured scalability, reliability, and better data management. Still, there are some limitations in MAC address filtering, which can be figured out with the machine learning approach.

VI. CONCLUSION

This is a smart solution to control the crowd by detecting crowd density without using traditional methods. This system can be used in a general environment where the problem of overcrowding is frequently encountered. This research relies on the design and development of an Internet of Things (IoT) based crowd detecting and monitoring system. The proposed solution presents a secure, real-time, and reliable method of estimating people within a given range and maintaining their privacy. This method relies on the phenomenon of passive capture of wi-fi probe requests transmitted by digital devices that are nearby. The collected data is processed on an Azure server and visualized via a web application dashboard. (As a future improvement, we can deploy it to mobile devices also.) By using a dashboard, users can directly access the web application and get an idea about the crowd density in that specific place. Because of this method, we can eliminate the demand for expensive equipment for image processing-based systems. We can use this system in different locations such as universities, offices, shopping malls, and transport hubs to improve the user experience, safety, and operational efficiency. This is a low-cost and very lightweight model, so it can be placed in a tiny space. The dynamic of the system's ability to capture and process data from the crowd, not only to monitor but also to process long-term, allows for both the immediate monitoring of the system, but also for predictive analytics based on the long-term data collection and analysis of such events. By observing the trend of congestion over time, the system can predict peak periods of congestion and possible crowding situations. This predictive ability enables better decision-making by the administrators and users to plan their activities more effectively and reduce waiting time, thereby enhancing space utilization.

However, the study indicates some limitations that need to be worked on in the future. Network coverage, interference, and device behavior directly affect the effectiveness of the Wi-Fi-based detection system. Those limitations can be addressed by going to a hybrid approach where this Wi-Fi probe request-based system is integrated with other computer vision-based systems.

Implementing machine learning algorithms could further increase the system's predictive capacity, enabling it to detect

patterns in people's behaviors and adjust its responses to evolving conditions.

In conclusion, the proposed IoT-based crowd monitoring system is a promising, cost-effective, and privacy-respecting solution to one of the most pressing issues of modern public management, overcrowding. It breaks the crucible between the traditional methods of manual or camera-based monitoring techniques and innovative and data data-driven analytics. The paper illustrates how the integration of Internet of Things (IoT), wireless communication, and intelligent data visualization techniques can help create more flexible, responsive, and sustainable environments. Ultimately, this system constitutes a step in the right direction of machine learning and creating more intelligent, safe, and efficient spaces that will help shape people and thus, better communities in a world where we are increasingly connected.

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REFERENCES

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